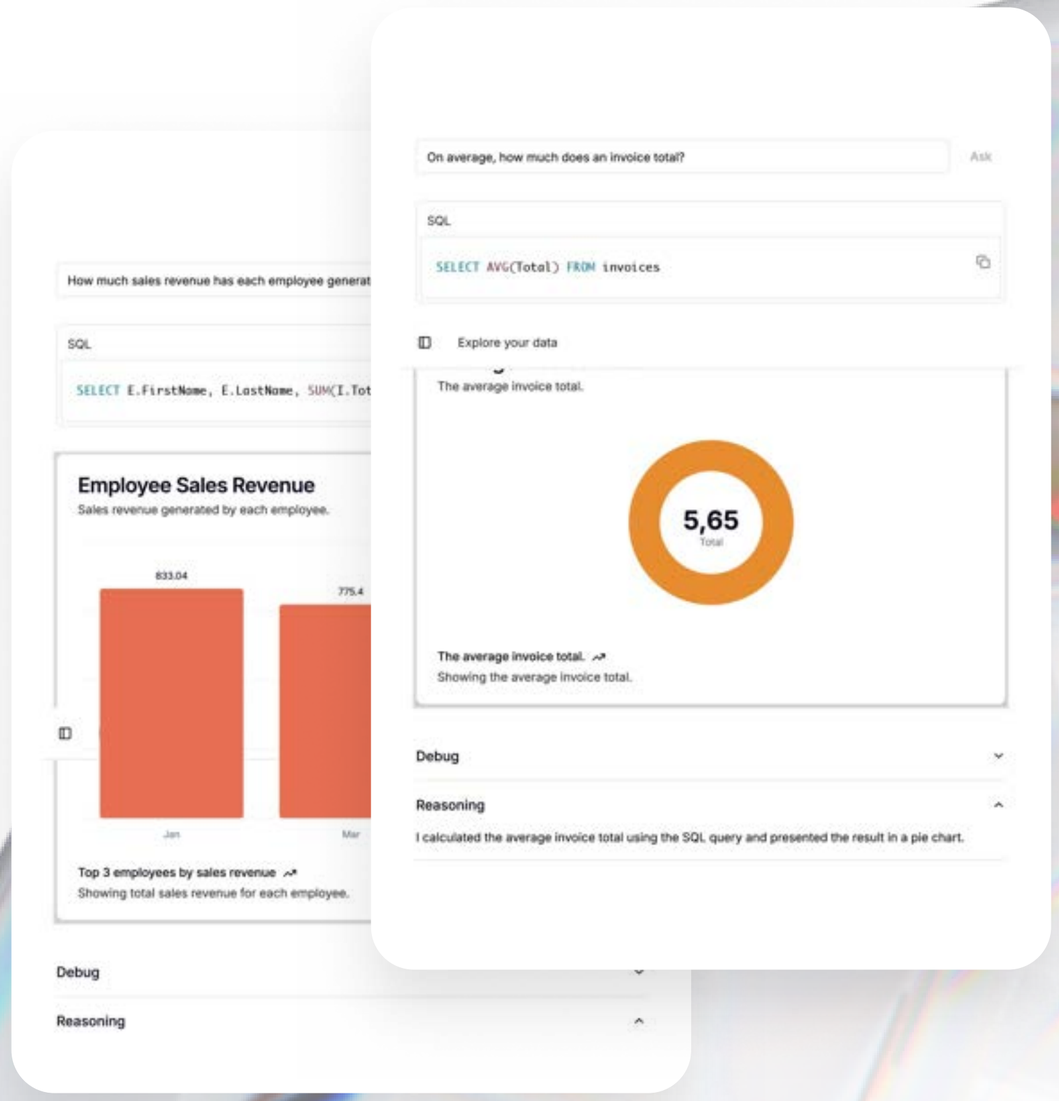
Conversational AI Assistant for Smarter Database Interaction

PoC | Architecture | AI-native engineering | SaaS deployment | QA testing

“Talk to Your Data” is a conversational AI assistant that showcases how GenAI can transform database interaction. his agent-based solution leverages LLMs and modern AI tools to translate user questions into SQL, execute them securely, visualize the results, and explain the reasoning behind each answer.



TECHNOLOGY STACK

Next.js | LangGraph.js | LangChain | OpenAI GPT-4o-mini | Anthropic Claude 3.7 Sonnet | Gemini Flash Lite 2.0 | LangChain SqlToolkit | LangChain SQLDatabase | LangChain Hub prompt templates | Zod

OUTCOMES

- 3 weeks - from concept to fully functional PoC
- 4 SQL tools - for reliable query execution and validation
- Delivered a conversational AI assistant for natural language data queries, visualization, and transparent explanations
- Empowered business and technical teams to explore data, build dashboards, and make decisions faster and with confidence